Real Numbers

The big idea

Every whole number greater than 1 is built from **prime numbers**, much like every word is built from letters. This chapter uses that one idea to find HCF and LCM quickly and to prove that numbers like √2 can never be written as a fraction.

REAL NUMBERS

Real numbers = rational numbers (can be written as p/q, q ≠ 0) + irrational numbers (cannot). Every point on the number line is a real number.

The Fundamental Theorem of Arithmetic

Every composite number can be written as a product of primes, and this factorisation is unique apart from the order of the factors.

For example, 3600 = $2^4 \times 3^2 \times 5^2$. However you start breaking 3600 down (36 × 100, or 60 × 60, or 2 × 1800…), you always end with four 2s, two 3s and two 5s.

PRIME FACTORISE 4410

Divide by the smallest prime that works, again and again:
4410 → 2 × 2205 → 2205 = 3 × 735 → 735 = 3 × 245 → 245 = 5 × 49 → 49 = 7 × 7.
So **4410 = $2 \times 3^2 \times 5 \times 7^2$**.

HCF and LCM using prime factors

Write each number as a product of primes, then:

- **HCF** = product of the **smallest** power of each **common** prime.
- **LCM** = product of the **greatest** power of **every** prime that appears.

FIND THE HCF AND LCM OF 96 AND 404

$96 = 2^5 \times 3$ and $404 = 2^2 \times 101$.

Common prime: 2, smallest power 2^2. So **HCF = 4**.

LCM = $2^5 \times 3 \times 101$ = **9696**.

Check: HCF × LCM = 4 × 9696 = 38784 = 96 × 404 ✓

FOR ANY TWO POSITIVE INTEGERS A AND B

$$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$$

THIS PRODUCT RULE IS ONLY FOR TWO NUMBERS

For three numbers, HCF × LCM is **not** equal to the product of the numbers. Use prime factorisation instead.

Why does 6^n never end in 0?

A number ends in 0 only if it is divisible by 10 = 2 × 5, so its prime factorisation must contain 5. But $6^n = 2^n \times 3^n$. By uniqueness of factorisation there is no 5 anywhere, so **6^n can never end in 0**. This is a typical 2-mark board question.

Proving √2 is irrational

We use **proof by contradiction**: assume the opposite and show it leads to something impossible.

1. Suppose √2 is rational, so √2 = a/b where a and b are co-prime (no common factor other than 1).
2. Squaring: $2 = a^2/b^2$, so **$a^2 = 2b^2$**. So a^2 is even, which means **a is even**. Write a = 2c.

3. Then $4c^2 = 2b^2$, so **$b^2 = 2c^2$**, which means **b is even** too.
4. Now a and b are both even, so they share the factor 2. This contradicts "co-prime".
5. So our assumption was wrong: **$\sqrt{2}$ is irrational**.

The same method proves $\sqrt{3}$ and $\sqrt{5}$ are irrational (replace 2 by 3 or 5 and use "if p divides a^2, then p divides a" for a prime p).

BUILDING ON A KNOWN IRRATIONAL

To show $5 - \sqrt{3}$ is irrational, assume $5 - \sqrt{3} = a/b$. Then $\sqrt{3} = 5 - a/b$, which is rational, a contradiction because $\sqrt{3}$ is irrational. The sum, difference, product or quotient of a non-zero rational and an irrational number is irrational.

Quick recap

- Prime factorisation is unique (Fundamental Theorem of Arithmetic).
- HCF: smallest powers of common primes. LCM: greatest powers of all primes.
- HCF × LCM = product, for two numbers only.
- Proof by contradiction shows $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ are irrational.

Practice questions

1. The prime factorisation of 140 is
   (a) $2 \times 7 \times 10$ (b) $2^2 \times 5 \times 7$ (c) $2 \times 5 \times 14$ (d) $2^3 \times 5 \times 7$
2. If HCF(26, 91) = 13, then LCM(26, 91) is
   (a) 91 (b) 2366 (c) 182 (d) 13
3. Which of these is irrational?
   (a) $\sqrt{16}$ (b) 0.375 (c) 22/7 (d) $3 + \sqrt{5}$
4. Why can 4^n never end with the digit 0?
   (a) Because 4 is even (b) Because its prime factorisation has no 5 (c) Because 4^n is always a perfect square (d) Because n is a natural number
5. HCF of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$ is
   (a) $2^2 \times 3^2 = 36$ (b) $2^3 \times 3^3 = 216$ (c) $2 \times 3 = 6$ (d) $2^3 \times 3^3 \times 5 \times 7$
6. In the proof that $\sqrt{2}$ is irrational, we start by assuming $\sqrt{2} = a/b$ where a and b are
   (a) both even (b) both prime (c) co-prime (d) equal

Answers: 1. b 2. c 3. d 4. b 5. a 6. c